

**Notes**

TEXTBOOK (THE WORLD AROUND US)

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6.1 INTRODUCTION

Textbooks are foundational tools in education, serving as structured resources to support teaching and learning. The pedagogical intent of a textbook reflects its purpose in guiding learners towards specific educational outcomes. Textbooks like 'Our Wondrous World' (class 3-5) are important educational tools that provide structured, curriculum-aligned content to foster learning in the early years. It ensures consistency in delivering needed knowledge, concepts, and

skills, making difficult ideas accessible through age-appropriate language and examples. Specifically, textbooks connect children's lived experiences to broader environmental and social themes, promoting curiosity, critical thinking, and values like empathy and sustainability. By including engaging stories, visuals, and activities, it supports holistic development, encourages active learning, and serves as a reliable resource for teachers and learners, ensuring equitable access to quality education across diverse contexts.

The 'Our Wondrous World' (Class 3-5) textbook plays a crucial role in shaping young learners' understanding of their environment, society, and the natural world. Designed primarily for elementary students, it serves as an engaging introduction to social studies and environmental education, fostering curiosity about the world they live in. The textbook covers a wide range of topics, including geography; history, culture, and environmental awareness, helping children develop a holistic perspective on their surroundings. By presenting concepts in an age-appropriate, interactive manner, it encourages critical thinking, empathy, and a sense of responsibility towards the planet and its diverse communities. Furthermore, it lays the foundation for global citizenship by promoting awareness of interconnectedness and sustainability, making it an essential resource for nurturing informed and conscientious individuals.

The pedagogical intent of the NCERT textbook 'Our Wondrous World' (Class 3-5) for the curricular area, 'The World Around Us' (Class 3-5), is to promote curiosity and build foundational knowledge in environmental and social studies through an engaging, child-centric approach. The textbooks incorporate innovative pedagogical strategies, to encourage active exploration, critical thinking, and experiential learning. By integrating relatable themes like playgrounds, gardens, and community life, it seeks to make learning meaningful and inclusive, helping young learners develop an understanding of their surroundings while nurturing values like empathy and environmental awareness.

This chapter explores the key pedagogical intentions behind textbooks, their principles for textbook design, and their role in the learning process. It further explains the development of learning objectives and selection of the learning experiences.



6.2 LEARNING OUTCOMES

After completion of this chapter, the learner

- identifies the pedagogical intent of textbooks;
- uses the textbook effectively;
- frames learning objectives clearly;



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- selects activities that promote active learning, critical thinking, and interdisciplinary connections;
- identifies learning experiences that align with curriculum goals and children's needs;
- evaluates the effectiveness of chosen learning experiences; and
- writes a specific lesson plan.

6.3 KNOWING A TEXTBOOK

What is a Textbook?

A textbook is a structured, authoritative book designed for educational purposes, typically aligned with a specific curriculum or course. It presents organized content-text, diagrams, examples, and exercises to teach knowledge or skills in a particular subject, such as math, history, or science. Textbooks aim to facilitate learning by providing clear explanations, logical progression, and tools like practice questions to support learners and teachers in achieving educational goals.

What are the Pedagogical Intents of a Textbook?

A textbook's objectives and teaching learning practices that are included into its content and structure are referred to as its pedagogical intent. The textbook offers progressive learning by providing age appropriate content that is intended to provide particular knowledge, abilities, and skills. Textbooks are designed to help teachers in providing effective instruction to children, to convey knowledge, support curriculum goals, promote understanding, encourage active learning, and scaffold learning. The textbooks have several pedagogical intents, such as:

➤ Providing Structured Knowledge

A textbook has defined goals and objectives, logical arrangement of subject content, aids and support for diverse needs, activities for implementation and practice, and assessment and evaluation which ultimately help the teacher in effective delivery of information. When the content is presented in such a structured manner it becomes easy for the learners to learn, retain and apply the knowledge in their real life.

➤ Promoting Active Learning

Textbooks encourage active learning by including elements and techniques that motivate children to absorb, apply, and internalize concepts instead of just passively memorizing them.

They aim to actively engage children by incorporating elements like interactive activities, practice questions and exercises, case studies, and integration of visuals and multimedia.

➤ **Catering to Diverse Learners**

Textbooks ensure accessibility and engagement of diversified learners by integrating different approaches that meet a range of learning demands, styles, and abilities. They foster an inclusive and productive learning environment by including these components, which ensure that learners with varying abilities, interest and backgrounds may access, interact with, and gain from the content. Textbooks should have clear and wide-ranging explanations, scaffolded content, differentiated instruction, multiple modalities, and cultural and contextual relevance so that varied learners can be benefited.

➤ **Encouraging Critical Thinking**

Textbooks should encourage critical thinking by including different methods that encourage learners to understand, analyze, evaluate, and apply knowledge beyond rote memorization. By including these elements, textbooks shift learning from passive absorption to active engagement, prompting children to connect knowledge to real life experiences. Textbooks should include features like open-ended questions, case studies and scenarios, comparative analysis, problem-based learning, reflective prompts, data interpretation, and extension activities so that the learner's critical thinking ability is enhanced.



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CHECK YOUR PROGRESS 6.1

1. Reflect on the textbook 'The World Around Us' of class 3, and enlist what features (e.g., summaries, questions, or visuals) helped you learn better?

How Can the Teachers Use the TWAU Textbook Effectively?

Textbooks play a very important role in providing knowledge to the children. To utilize them to their maximum potential, it becomes very essential that the teacher should know how to use the textbooks effectively. Here's a concise guide to help teachers.

Understand the Textbook's Structure and Objectives

- **Review the Structure:** Teachers should first review the structure of the textbook; analyze the table of contents, chapter objectives, glossary, and sequence of content. They should note repeated themes (e.g., environmental conservation, community roles).

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- **Identify Key Features:** Teachers should also identify key features like discussion questions, note activities, diagrams, maps, or case studies included in the textbook 'Our Wondrous World' (Class 3-5).
- **Align with Learning Goals:** Teachers should map the textbook content to specific learning goals or standards (observation, classification, mapping of data, demonstration of different skills) to ensure relevance.

**ACTIVITY 6.1: Textbook Exploration**

1. Open the textbook 'Our Wondrous World' (Class-3) and select any chapter (e.g., "Going to the Mela").
2. List the following:
 - Key topics covered
 - Types of activities or questions provided
 - Any visuals (maps, charts, pictures) and their purpose
3. Write a brief paragraph aligning the chapter's objectives to one or two curriculum competencies (observation, classification, mapping of data, demonstration of different skills).

**CHECK YOUR PROGRESS 6.2**

1. Which features of the textbook are you using to support your teaching goals?

Make Lessons Interactive and Engaging

To engage children, and help them move beyond rote learning and use the textbook as an effective tool for active learning:

Include Hands on Activities: Use the suggested activities in the textbook (e.g., nature walk, the solar system, or experiments) to make learning hands-on. For example, if the chapter discusses the solar system, you can develop an activity by assigning the role of different planets to the children.

Incorporate Visuals and Media: You can supplement the textbook by including different pictures, real life objects. If facilities are available you can make it more interesting by incorporating videos, games etc.

Encourage Inquiry-Based Learning: Use the textbook's questions or case studies as prompts for discussion, role-plays, and homework. For example, ask the children to discuss environmental issues like 'pollution' mentioned in the textbook with their parents.

**ACTIVITY 6.2: Lesson Plan Draft**

1. Choose a topic from 'Our Wondrous World' (class 3) (e.g., "Living in Harmony").
2. Design a 30-minute lesson incorporating:
 - A warm-up activity (e.g., a game, role play or discussion to generate interest).
 - A textbook-based activity (e.g., reading a section and discussing key terms).
 - An interactive group task (e.g., designing a poster).
3. Include at least one question from the textbook to guide the discussion.

*Notes***CHECK YOUR PROGRESS 6.3**

1. How did your lesson plan make the textbook content more engaging? What additional resources would you like to use in the future to enhance engagement and interest of learners?

Differentiated Instruction

Learners have varied learning styles and abilities so it's your duty as a teacher to adapt the textbook, 'Our Wondrous World' (class 3-5), according to the needs of the diverse learner. The following are some techniques to adapt the textbook for providing differentiated instruction:

Cater to Diverse Learners: Adapt the textbook content for varying skill levels. For struggling learners, try to fill the learning gaps by explaining the topic from the basics, and highlighting the key points or use graphic organizers. For advanced learners, assign extension activities like studying a topic in depth.

Incorporate Group Work: The whole class should be divided into heterogeneous groups, work should be assigned according to abilities, then learners asked to present summaries, create posters, or lead discussions related to the lesson. This fosters peer learning, collaboration and deeper understanding among learners.

Use Real-World Connections: Relate the textbook topics to children's lives. For example, when discussing community helpers, ask the children to talk about local workers in school, in the neighborhood or share personal experiences.

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**ACTIVITY 6.3: Differentiation Plan**

1. Select a chapter from 'Our Wondrous World' (class-4) (e.g., "Growing up With the Nature").
2. Create a plan with:
 - One adaptation for struggling learners (e.g., simplified reading guide).
 - One extension activity for advanced learners (e.g., deep analysis of different type of flowers).
 - One group activity to encourage collaboration (e.g., poster making).
3. Enlist methods to connect the topic to children's lives (e.g., gardening).

**CHECK YOUR PROGRESS 6.4**

1. How will you ensure involvement of all the children while using the textbook 'Our Wondrous World' (Class 3-5)?

Assessing Learning and Real-World Connections

Assessment is one of the most important aspects of the teaching and learning process; here you have to ensure the achievement of the learning outcomes. As the children learn in different ways, assessment should also be designed accordingly. Following are some of the ways which can be very helpful in assessing the children.

Use Textbook Questions Strategically: Utilize end-of-chapter questions for formative assessments, quizzes, or class discussions. Modify questions to encourage critical thinking (e.g., "What would happen if..." instead of recall-based questions).

Encourage Reflection: Ask the children to write or discuss how the textbook's topics are applicable to their daily life.

Create Customised Assessments: Design activities, projects based on textbook themes, such as creating a model of different phases of the moon or a campaign to reduce pollution.

Connect to Current Events: Relate topics to usual phenomena e.g., reason of drought and flood.

ACTIVITY 6.4: Assessment Design

1. Choose a chapter from the textbook 'Our Wondrous World' class-3 (e.g., "Food We Eat").

2. Design two assessments:
 - A formative assessment using a textbook question (e.g., a short quiz).
 - A project-based assessment (e.g., children enlist different foods.)
3. Organize a discussion in the classroom connecting the topic to children's daily life (e.g., "How does eating a variety of foods help in gaining good health?").



CHECK YOUR PROGRESS 6.5

1. How do your teaching methods encourage children to apply textbook knowledge to real life?

Integrate Technology and Real-Time Resources

Integrating technology into teaching TWAU for young learners can make the teaching learning process interactive, engaging, and effective. The aim is to use age-appropriate techniques and tools that foster interest, curiosity, improve understanding of environmental concepts, and raise active participation. Following are some of the ways to integrate technology, keep the subject matter simple and align it with the cognitive level of young learners:

Educational Videos and Animations

You can utilize different YouTube Channels like SciShow Kids or BrainPOP Jr. etc., that offer short animated videos tailored to explain difficult topics and curated playlists from platforms like PBS Learning Media for kid-friendly TWAU content.

Interactive Apps and Games:

Engage children with fun, interactive tools to learn about this curricular area. Use apps like Toca Nature or Eco Kids (available on iOS/Android) where they can explore virtual ecosystems, plant trees, or learn about recycling through games. Platforms like National Geographic Kids or NASA Climate Kids offer interactive games, quizzes, and videos on topics like weather, wildlife, and pollution.

Virtual Field Trips:

Expose children to real-world environments like forests, oceans, or recycling centers without leaving the classroom. For this you can use Google Earth and Websites like Discovery Education or The Nature Conservancy.



ACTIVITY 6.5: Technology Integration



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1. Select a chapter from the textbook 'Our Wondrous World' (Class-5) (e.g., Water – The Essence of Life).
2. Identify one online resource (e.g., a simulation tool or video) to supplement the textbook.
3. Plan a 10-minute activity using this resource (e.g., children learn to differentiate drinkable or non-drinkable water).

**CHECK YOUR PROGRESS 6.6**

1. How does technology make the textbook content easy and interesting for young learners? Support your answer with some examples.

Foster Critical Thinking and Application

Connect to Current Events: Relate the textbook topics to current or local issues. For example, if the book covers water conservation, discuss rainwater harvesting, recent droughts and floods.

Encourage Problem-Solving: Use case studies or scenarios from the textbook to have children propose solutions to real-world problems, like pollution control or waste management.

Promote Cross-Disciplinary Learning: Link TWAU to other subjects, such as language, mathematics, art education and physical education.

Putting it all Together

Create a comprehensive lesson plan of a lesson from the textbook 'Our Wondrous World' classes (3-5).

**ACTIVITY 6.6: Comprehensive Lesson Plan**

Using the strategies from this module, create a full lesson plan for one chapter of 'Our Wondrous World'. Include:

- **Objective:** One clear learning outcome.
- **Warm-Up:** A 5-minute activity to engage children.
- **Main Activity:** A 20-minute task using textbook content (e.g., reading, discussion, or hands-on activity).
- **Differentiation:** One strategy each for struggling and advanced learners.
- **Technology/Real-World Connection:** One tech tool or current event link.
- **Assessment:** One formative or summative assessment.

**CHECK YOUR PROGRESS 6.7**

1. How does your lesson plan balance textbook use with children's engagement and real-world relevance? Write your answer after trying your lesson plan in the classroom.

6.4 DEVELOPING LEARNING OBJECTIVES (LOs)

Learning objectives are clear statements that describe what learners should know, understand, or be able to do by the end of a lesson. It helps in lesson planning, selecting teaching methods, learning experiences and assessment methods. Learning objectives should be specific, measurable, achievable, relevant, and time-bound.

6.4.1 Steps to Develop Learning Objectives

Teachers should follow the following steps to develop effective learning objectives.

Step 1: Identify the Content of the Lesson

The teacher should deeply analyze the lesson, find out the concept, definition and important points given in the content e.g. in a lesson on “Water-The Essence of Life” from textbook ‘Our Wondrous World’ (Class-5), the key concept might be to understand the importance of water conservation.

Step 2: Consider Your Students

Consider the age, previous knowledge, and learning requirements of your students. While older students can do more complicated tasks like analysis or application, younger learners (Class 3–5) can require simpler objectives centered on fundamental principles e.g. Class 3 students, focus on identifying ways to conserve water, while for Class 5, children could compare water conservation methods across different cultures.

Step 3: Use Action Verbs (Bloom's Taxonomy)

For the textbook ‘Our Wondrous World’ (Class 3–5), verbs like enlist, describe, explain, etc. are often appropriate for lessons on TWAU. Choose action verbs that reflect the cognitive level you want children to achieve, based on revised Bloom's Taxonomy:

- **Remembering:** List, identify, describe, recall
- **Understanding:** Explain, summarize, compare



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- **Applying:** Use, demonstrate, solve
- **Analyzing:** Compare, analyze, categorize
- **Evaluating:** Assess, justify, critique
- **Creating:** Design, create, propose, develop

Example: Students will list three ways to conserve water in their locality.

Step 4: Align with Curricular Goals/Competencies

Make sure the curricular goals correspond with the competencies mentioned in NCFSE for TWAU e.g. If a competency requires learners to “understand the impact of human activities on the environment,” your objective could be: “Children will explain how pollution affects local water sources.”

Step 5: Make Objectives Measurable

Objectives should be specific so that their measurement can be done easily. Include criteria to assess whether children have met the objectives. Use phrases like list three examples, create a poster, or explain in a short paragraph e.g. learners will create a poster illustrating two methods of water conservation.

Step 6: Write the Objective of Your Class

Combine all these elements into a concise statement:

Who: The children of Class 3

What: The skill or knowledge to be acquired (use an action verb).

How: The context or method (e.g., through a discussion, using a map).

Why: The purpose or relevance (optional).

Example: By the end of the lesson, “Learners will be able to describe three water conservation methods in a group discussion” from the textbook ‘Our Wondrous World’ Class-5.



ACTIVITY 6.7: Writing Your Own Learning Objectives

1. **Choose a Chapter:** Select a chapter from textbook ‘Our Wondrous World’ (class 3-5).
2. **Identify Key Concepts:** List 2–3 main ideas or skills from the chapter.
3. **Select a Grade Level:** Decide the age group (e.g., class 3–5).
4. **Write Two Objectives:** Create one objective for a lower cognitive level (e.g., remembering or understanding) and one for a higher level (e.g., applying or creating).

5. **Reflect:** Check if your objectives are specific, measurable, achievable, relevant, and time-bound.

6.5 SELECTING LEARNING EXPERIENCES

Selection of learning experiences is one of the most important phases of the teaching learning process. You should select the learning experiences on the basis of the learner's age, interest, and intelligence etc. It should be engaging for the learners, so that their attention can be attained by the teachers. To select the appropriate learning experiences teachers should analyze the textbook content, examples, and activities and try to identify the core themes of the lesson. After identifying the core theme you as a teacher should try to contextualize it and select activities, stories, examples, games etc. from the surroundings of the learner, so that they can relate to it and can understand the theme better. You should focus on those learning experiences that help to inculcate or develop competencies among the learner as suggested by NCFSE.



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ACTIVITY 6.8

1. List three key themes from any lesson of the textbook 'Our Wondrous World' (Class 3-5) that relate with your children's lives.
2. How do these learning experiences align with your curriculum competencies?

6.5.1 Principles for Selecting Learning Experiences

Effective learning experiences are purposeful, engaging, and inclusive. You can use the following principles while selecting learning experiences for the learners:

Relevance: Choose activities that connect textbook content to the children's lives or current events (e.g., local environmental issues).

Active Engagement: Prioritize hands-on, interactive tasks over passive learning (e.g., role play, games and activities).

Differentiation: Select learning experiences that address diverse learning needs of learners, including visual, auditory, kinesthetic and cognitive.

Inclusive: Learning experience should not exclude the learners on the basis of cultural, regional, intellectual differences. Selection of learning experiences should be made in such a manner that all learners feel connected to it and benefit from it.

Interdisciplinary Connections: The teacher should integrate subjects like Language, Science, Mathematics, Art, Geography, and History according to their level to deepen understanding.

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Assessment Alignment: Ensure activities allow children to demonstrate understanding in measurable ways.

Social and Collaborative Value: Activities ought to promote collaboration, dialogue, and consideration for the opinions of others.

Example: A chapter from the textbook ‘Our Wondrous World’ Class-4 “Growing Up with Nature”, a teacher might select a visit to garden, lake, river and different types of natural places to show their importance, teacher can give an assignment to them to do with the help of their parents on “Impact of Nature on Human Life”.

6.5.2 Steps of Selecting Learning Experiences

Follow this step-by-step process to select appropriate learning experiences:

Identify Learning Competencies: Determine what learners should know or do after the lesson (e.g., “Understand the water cycle”).

Assess Student Needs: Consider age, learning styles, and prior knowledge. For example, younger children may need hands-on activities, while older children can handle complex tasks.

Match Activities to Content: Select experiences that directly relate to the book’s themes. For instance, a chapter from ‘Our Wondrous World’ Class-3 “Going to the Mela” might relate with their nearby Melas.

Plan for Inclusivity: Ensure activities accommodate diverse learners i.e. selecting those learning experiences which suit all learners.

Evaluate Feasibility: Check availability of resources, time, and space. For example, a field trip may require advance planning.

Design Assessments: Create rubrics or checklists to measure children’s outcomes (e.g., accuracy of a task or participation in discussions).

6.5.3 Evaluating Learning Experiences

To ensure effectiveness, the teacher should evaluate activities based on:

Student Engagement: Are the children actively participating and showing interest?

Learning Outcomes: Do the children demonstrate understanding of concepts?

Inclusivity: Are all children, including those with special needs, able to participate?

Feedback: Collect children and peer feedback to refine activities.

Example Evaluation Tool:

- Create a checklist: “Did the activity align with the book’s theme?”
- Did it accommodate diverse learners?
- Did it meet the learning goals?
- Observe children’s interactions during the activity and note engagement levels.
- Create Rubrics: Scoring systems that characterize performance levels across understanding, analysis, effort, and originality criteria.

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Practical Tips for Implementation

Start Small: Begin with one or two new activities per chapter to avoid overwhelming the children.

Integrate Student Choice: Allow children to choose project topics (e.g., a specific country or ecosystem) to motivate them and to encourage decision making.

Connect to Local Context: Relate book themes to local issues (e.g., a nearby river for a water conservation lesson).

Collaborate with Colleagues: Share activity ideas with other teachers to refine and diversify approaches.



CHECK YOUR PROGRESS 6.8

1. How will you select learning experiences for the learners?

6.6 PROPOSED CURRICULAR GOALS, COMPETENCIES AND LEARNING OUTCOMES (NCF-2023) ‘OUR WONDROUS WORLD’

Curricular Goals (CG) 1: Explores and engages with the natural and socio-cultural environment in their surroundings.

Competencies (C)

- **C-1.1** Observes and identifies the natural (insects, plants, birds, animals, geographical features, sun and moon, stars, planets, natural resources) and social (houses, relationships) components in their immediate environment.

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- **C-1.2** Describes relationships (including between humans and animals/nature) and traditions (art forms, celebrations, festivals) in the family and community.
- **C-1.3** Asks questions and makes predictions about simple patterns (season change, food chain, phases of the moon, movement of stars and planets, shapes of trees, plants, leaves, and flowers, rituals, celebrations) observed in the immediate environment.
- **C-1.4** Explains the functioning of local institutions (family, school, bank/post office, market, and *Panchayat*) in different forms (story, drawing, tabulating data, reports), and analyses their roles.
- **C-1.5** Uses local materials to create simple objects (family tree, envelopes, origami animals) on their own for display or use in classroom processes.

CG 2: Understands the interdependence in their environment through observation and experiences, developing the basis for appreciation of the idea of ‘Vasudhaiva Kutumbakam’.

Competencies:

- **C-2.1** Identifies natural and human-made systems that support their lives (water supply, water cycle, river flow systems, seasons, life cycle of plants and animals, food, household items, transport, communication, electricity in the home).
- **C-2.2** Describes the relationship between the natural environment and cultural practices in their immediate environment (nature of work, food, festivals, traditions).
- **C-2.3** Connects changes in the environment and the lives of their family and community, as communicated by elders and through local stories (changes in occupation, food habits, resources, celebrations, communication).

CG: 3 Explains how to ensure the safety of self and others in different (normal, as well as, emergency) situations.

Competencies:

- **C-3.1** Describes the basic safety needs and protection (health and hygiene, food, water, shelter, precautions, awareness of emergency situations, abuse, and unsafe situations) of humans, birds, and animals.
- **C-3.2** Discusses how to prepare for emergency situations (smoke, fire, small injuries, burns, electrical safety, unseasonal rains, fallen trees) based on discussions with family and community, or personal experiences.

- **C-3.3** Develops simple labels and slogans, and participates in roleplay on safety and protection in the local environment to be displayed/done in school and locality.

CG 4: Develops sensitivity towards social and natural environment.

Competencies:

- **C-4.1** Observes and describes diversity among plants, and birds and animals in their immediate environment (shape, sounds, food habits, growth, habitat).
- **C-4.2** Observes and describes cultural diversity in their immediate environment (food, clothing, games, different seasons, festivals related to harvest and sowing).
- **C-4.3** Describes usage of natural resources in their immediate environment.
- **C-4.4** Demonstrates how natural resources can be shared, maintained, and conserved (trees, use of rainwater, benefits of millets).
- **C-4.5** Identifies needs of plants, birds, and animals, and how they can be supported (water, soil, food, care).
- **C-4.6** Identifies the needs of people in different situations, in terms of access to resources, equal opportunities, work distribution, and shelter.
- **C-4.7** Learns about basic social and behavioural norms, values, and dispositions that benefit our social and natural environments and that help our society function smoothly (using dustbins, standing in queues, conserving water, using public transportation, keeping one's environment clean, always helping others in need regardless of background).

CG 5: Develops the ability to read and interpret simple maps.

Competencies:

- **C-5.1** Explains a line drawing of their school, village, and ward.
- **C-5.2** Draws a sketch of their school, village, and ward using symbols and directions.
- **C-5.3** Reads simple maps of city, state, and country to identify natural and human-made features (well, lake, post office, school, hospital) with reference to symbols and directions.

CG 6: Uses data and information from various sources to investigate questions related to their immediate environment.

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Competencies:

- **C-6.1** Performs simple inquiry related to specific questions independently or in groups.
- **C-6.2** Presents observations and findings through different creative modes (drawing, diagram, poem, play, skit, oral and written expression).

CG 7: Gains foundational familiarity with basic concepts and methods from the natural sciences (life sciences, physical sciences, and earth and space sciences) and engineering.

Competencies:

- **C-7.1** Gains familiarity with using the scientific method in investigations, as well as, familiarity with other crosscutting concepts, such as, energy, matter, and systems that apply across the domains of science and engineering.
- **C-7.2** Gains familiarity with disciplinary core ideas in the natural sciences, as well as, in engineering, technology, and applications of science, which reflect the content that will be learned across subject areas in later Grades.

Learning Outcomes

Learning Outcomes are interim markers of learning achievement towards the attainment of competencies. They are defined based on the specifics of the socio-cultural contexts, the materials and resources available, and the contingencies of the classroom. In the following section a sample lesson plan is given in which learning outcomes are framed for a class.

6.7 SAMPLE LESSON PLAN (OUR WONDROUS WORLD)

A detailed sample lesson plan for Class 5 is given. You will be able to write your specific lesson plan after reading it.

Class: 5

Book: 'Our Wondrous World' Class-5

Unit-1: Life Around Us

Lesson-1: "Water-The Essence of Life"

Duration: 40 Minutes

Learning Aspects

- Importance of freshwater for survival (drinking, crops, daily activities)
- Forms of water: liquid (rain), solid (ice/snow), vapor (steam)
- Different forms of water (solid, liquid, gas)

Learning Outcomes/Objectives:

After completion of the lesson students will be able to:

- explain why water is essential for life;
- identify its sources (freshwater and saltwater); and
- describe the different forms of water (solid, liquid, gas).

Process Skills

- Observation
- Classification
- Communication
- Recording and interpreting data

Learning Resources:

- Visual aids: Map of India showing major rivers
- Activity materials:
 - Images of freshwater sources (printed or projected).
 - Glass and spoon
 - Steel glass, ice cubes

Opportunities for Open-ended Discovery:

- Children discover the different sources of water.
- Children explore different forms of water
- Open discussion: “Why do we need to conserve water?”

5E Instructional Model

Engage -Teacher start with the questions

1. How many of you like rain?
2. What would happen if there was no water on earth?
3. How do we get water for our uses?

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4. Where does rain come from?
5. So today I will tell you the story of rain and water.

Explore

- Begin with the chapter's story: Read aloud the opening scenario where Afreen and Jyoti watch raindrops and wonder about water's journey.
- The teacher shows pictures of the ocean, lake and river.
- The teacher also shows pictures of ice, steam and water.
- Show a short video clip (1-2 minutes) of rain or a river.

Explain

Teacher's Activity: Ask students which one is bigger among ocean, lake and river?

Student's Activity: Ocean

Teacher's Activity: Do the taste of Ocean water and lake water is same?

Student's Activity: Ocean water is salty

Teacher's Activity: Ask students if the ice and vapour contains water?

Student's Activity: Yes

Elaborate

Teacher's Activity: Teacher elaborates that oceans are big but their water is salty so we cannot drink it. We can use it to make the salt. The teacher takes a full glass of water and takes one spoonful of water. She then explains that if all the water on Earth were contained in this glass, the freshwater would be the same as the amount in a teaspoon. Teachers explain that the survival of humans, animals, and trees depends on fresh water. To create the salt, ocean water is required.

Teacher's Activity: The teacher perform an activity by using ice crystal and boiling water. The teacher takes few ice crystals in a glass. At the same time she brings another pot of water to a boil. As a result, ice crystal eventually begins to melt and water begins to turn into vapor. The teacher goes on to explain that boiling water turns it into vapor, melting ice turns it into water, and when we keep water in freezer it turns into ice crystal. Thus, water exists in three different forms: solid (ice), gas (vapor), and liquid (water).

Student's Activity: Participates in activity and ask questions.

EVALUATE

Teacher's Activity: Gives a short quiz or worksheet to match water taste with their source.

Student's Activity: Completes the quiz/worksheet; explains the forms of water in their own words.

Reflection

The teacher will write the reflection about the class. He will write how effectively the method chosen to teach, activities, examples or any other things are worked in the achievement of the learning outcomes.

**CHECK YOUR PROGRESS 6.7**

1. Write a lesson plan from any lesson of the textbook 'Our Wondrous World' (Class 3-5).

6.8 LET US SUM-UP

Textbooks like 'Our Wondrous World' serve as pivotal educational tools, providing structured, curriculum-aligned content that fosters foundational learning in young learners. Teachers should remember that textbooks serve as structured resources, guiding both teachers and learners through carefully curated content designed to meet pedagogical goals. Understanding a textbook's intent and structure prepares teachers to utilize it effectively for teaching, developing differentiated instructions, incorporating technology and fostering meaningful engagement with content. Developing clear and measurable learning objectives is critical for selecting teaching methods and assessing learners' progress. Teachers should always remember that learning objectives should align with curriculum goals and promote holistic development. Similarly, selecting appropriate learning experiences is important for creating engaging and impactful learning. Teachers should take care that learning experiences should be purposeful, engaging, contextual, inclusive, and aligned with desired learning goals. Teachers should know how to evaluate these learning experiences because evaluating these learning experiences ensures their effectiveness, offering teachers to improve methods based on learner outcomes and feedback. Teachers should always consider the proposed learning standards in NCF-2023 which emphasize a learner-centric approach, integrating flexibility and innovation to meet contemporary educational demands. Additionally, the proposed learning standards from the National Curriculum Framework (NCF-2023) provide a forward-looking vision, emphasizing competency-based education and learner-centric pedagogies. Together, these elements empower teachers to create dynamic, inclusive, and outcome-driven learning environments that prepare learners for future challenges.

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6.9 END EXERCISE

1. What is the primary purpose of a textbook in education?
2. How do textbooks cater to different learning styles?
3. Identify one feature of a textbook that promotes critical thinking.
4. How can you align a learning experience with both the book's themes and your curriculum standards?
5. What strategies can you use to differentiate activities for diverse learners?
6. How will you evaluate the success of a learning experience in your classroom?
7. Choose one theme from textbook 'Our Wondrous World' (Class 3-5) and design a brief learning experience using one of the above approaches. Reflect on how it engages your children.

6.10 REFERENCES

- Centre for Science and Environment. <https://www.cseindia.org>
- Edutopia. (2020). Project-Based Learning: A Resource Guide. Retrieved from <https://www.edutopia.org/project-based-learning>
- <https://www.education.gov.in>
- Indira Gandhi National Open University. (n.d.). Unit 11: Teaching-learning resources. School of Education, IGNOU. <https://www.ignou.ac.in>
- Ministry of Education. (2020). National Education Policy 2020. Government of India.
- Ministry of Education. *DIKSHA: Digital infrastructure for knowledge sharing*. <http://diksha.gov.in>
- National Curriculum Framework (NCF) 2023, NCERT. <https://ncert.nic.in>
- National Research Council. (2012). A Framework for K-12 Science Education: Practices, Crosscutting Concepts, and Core Ideas. The National Academies Press.
- NCERT. (2024). Our Wondrous World. A Textbook for Class III. The World Around Us.
- NCERT. (2025). Our Wondrous World. A Textbook for Class IV, The World Around Us.
- NCERT. (2025). Our Wondrous World. A Textbook for Class V, The World Around Us.

- TESS-India. (n.d.). Open educational resources for teacher education. The Open University. <http://www.tess-india.edu.in>
- Tomlinson, C. A. (2017). How to Differentiate Instruction in Academically Diverse Classrooms. ASCD.
- UNESCO. (2017). Education for Sustainable Development Goals: Learning Objectives. UNESCO Publishing.
- Wiggins, G., & McTighe, J. (2005). Understanding by Design. ASCD.

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